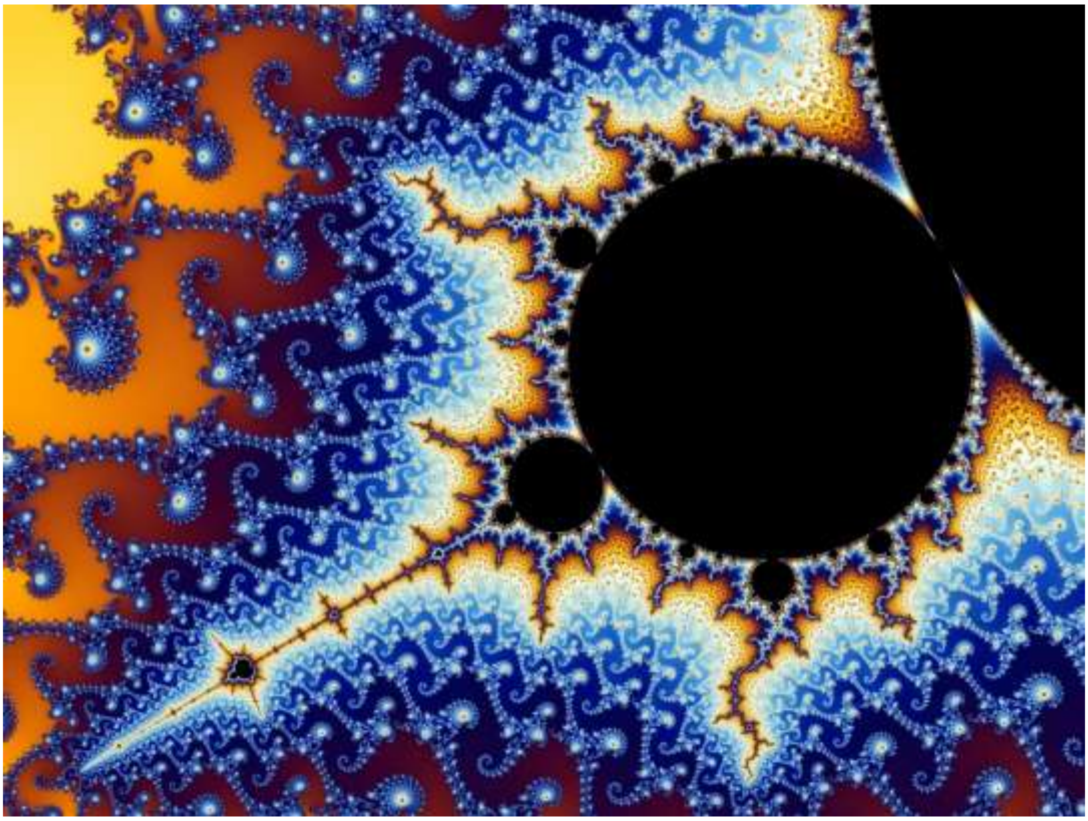
ST STITHIANS COLLEGE

Tour de Maths 2019

Leg 1

Question Paper - MEMO



The Mandelbrot set

5 Mark Questions

Question 1

Tasneem was going to buy eight sweets at the tuckshop but it turned out she was R7 short. Instead, she only buys seven sweets and is left with R5 to spare.

How much did a single sweet cost if all the sweets she bought cost the same?

MEMO

	Sweets	Money
Situation A	8	7 short
Situation B	7	5 extra
Difference	1	12 Rand

ANSWER: R12

Question 2

Which day of the week was it a million seconds ago?

MEMO

$$\frac{1000000\,\text{sec}}{60} = 16666,6\dot{6}\,\text{min}$$

$$\frac{16666,6\dot{6}\,\text{min}}{60} = 277,7\dot{7}\,\text{hours}$$

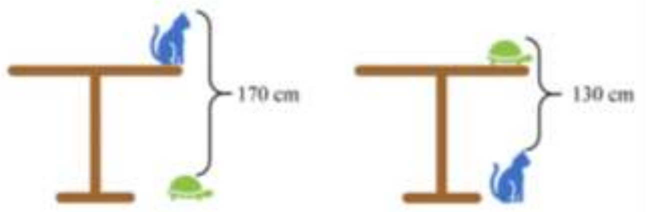
$$\frac{277,7\dot{7}\,hours}{24} = 11,5\dot{7}40\dot{7}\,\text{days}$$

therefore if today is Thursday, we go back 11 days we are on a Sunday

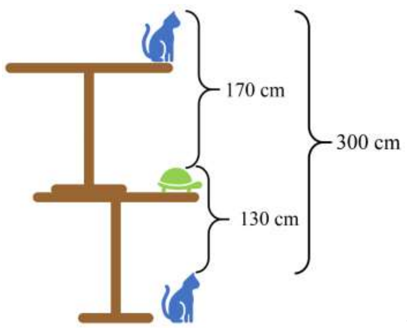
ANSWER: SUNDAY

Question 3

How tall is the table?



MEMO



So 1 table is 150cm

ANSWER: 150cm

Question 4

A clock with no numbers fell to the ground.

What time is it?

(a) 4:18 (b) 5:23
(c) 10:48 (d) 11:53

MEMO

The hour hand is $\frac{4}{5}$ way to next hour marker.

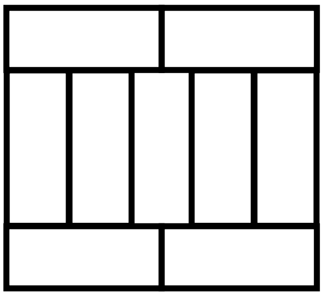
The minute hand has to be $\left(\frac{4}{5}\right).60 = 48$ mins past the hour. Therefore between "9" and "10", making the hour hand between "10" and "11".

Therefore 10:48.

ANSWER: C

Question 5

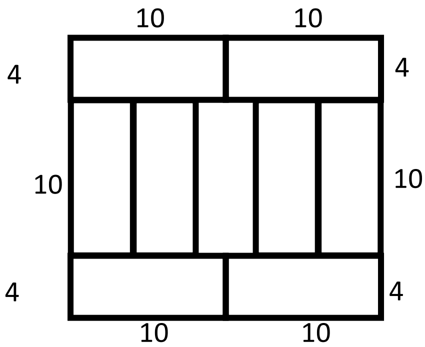
In the diagram, a large rectangle is made up of nine small identical rectangles whose longest sides are 10cm long.



What is the perimeter of the large rectangle?

MEMO

5 widths span 2 lengths therefore each width is $\frac{20}{5} = 4$



ANSWER: 76

Question 6

A hotel on an island in the Caribbean advertises using the slogan "350 days of sun every year!". According to the advert, what is the smallest number of days Rihanna has to stay at the hotel in 2019 to be certain of having two consecutive days of sun?

MEMO

$365 - 350 = 15$ non sunshine

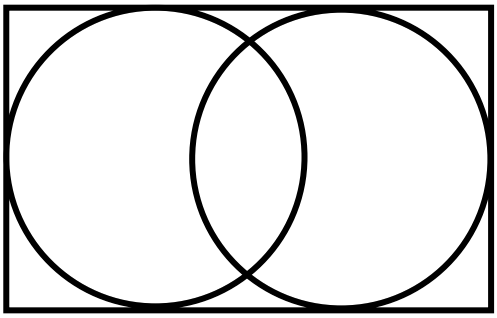
If we have 1 day of sun and then we wait maximum time for non sun we have a 16 day interleave. The minimum we would have to wait is two of those cycles therefore 32 days minimum.

ANSWER: 32

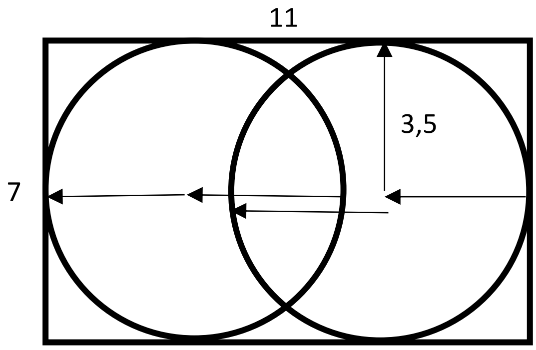
Question 7

The diagram shows a rectangle of dimensions 7×11 containing two circles, each touching three of the sides of the rectangle.

What is the distance between the centres of the two circles?



MEMO



the moved circle has shift 11 – 7 units from its original position. Therefore 4 units between centres

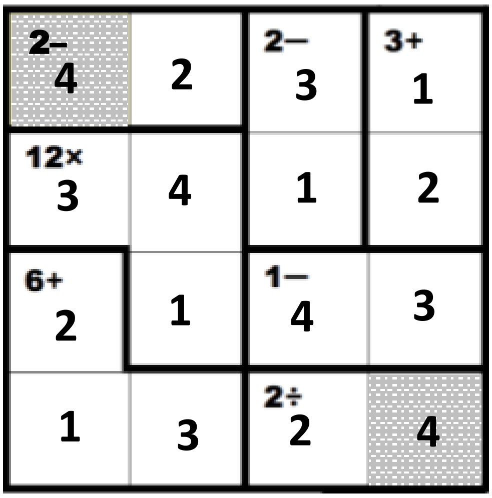
ANSWER: 4

Question 8

To solve a 4 x 4 KenKen puzzle, fill in the blank squares so that each row and each column contain only all of the digits 1, 2, 3 or 4. The heavy lines indicate areas (called cages) that contain groups of numbers that can be combined (in any order) to produce the result shown in the cage, with the indicated operation. For example, 12 $\times$ means you can multiply the values together to produce 12.

What is the sum of the numbers in the two shaded squares?

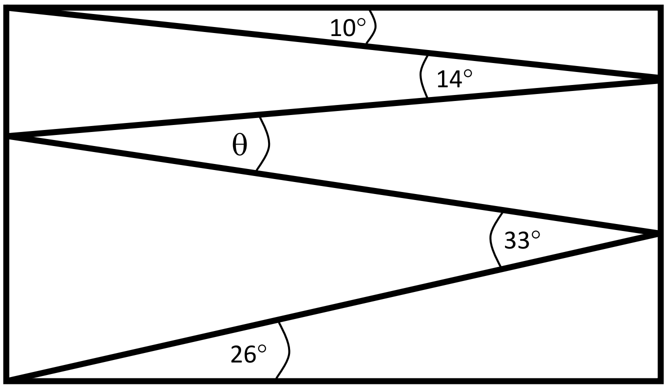
MEMO



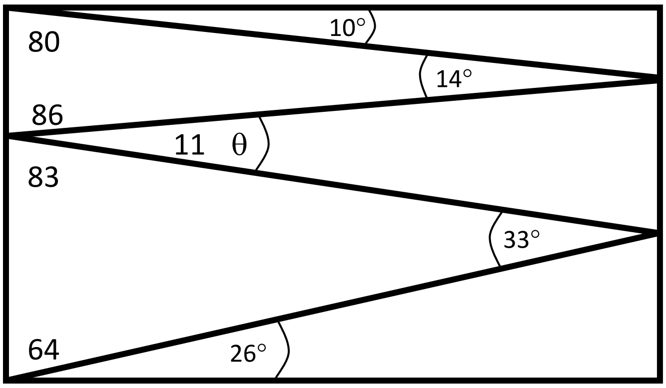
ANSWER: 8

Question 9

Naledi draws a zig-zag line inside a rectangle, creating angles of 10°, 14°, 33° and 26° as shown. What is the size of angle θ?



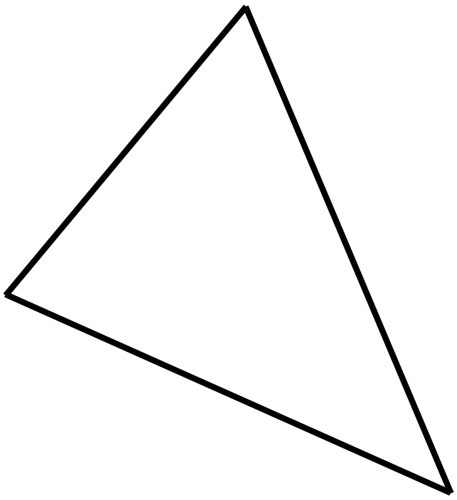
MEMO



ANSWER: 11

Question 10

The lengths of the sides of a triangle are x, 16 and 31, where x is the shortest side.
If the triangle is not isosceles, what is a possible value of x?



MEMO

By the triangle rule, x lies between 31 – 16 = 15 and 31 + 16 = 47. That is, we have $15 < x < 47$. But we are also given that x is the length of the shortest side of the triangle. So $x < 16$.Therefore we can grid in any number between 15 and 16.

ANSWER: 15 < x < 16

10 Mark Questions

Question 11

Determine natural number x so that (x^2+2065) is the square of a natural number (a perfect square)

MEMO

$$(x^2+2065)=\mathrm{n}^2$$
$$2065=n^2-x^2$$
$$2065=(n+x)(n-x)$$
$$(59)(35)=(n+x)(n-x)$$
$$(59)(35)=(47+12)(47-12)$$
$$n=47 \quad x=12$$

ANSWER: 12

Question 12

Four married couples: Agatha and John, Barbara and Kevin, Celine and Deon, and Daphne and Matthew were celebrating Matthew's birthday. Everybody was sitting at a round table in such a way that each lady was seated between two gentlemen and all the couples were separated. Agatha took her seat between Kevin and Matthew. Matthew sat to the right of Agatha. John was sitting next to Daphne.

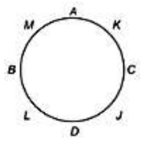
Who is sitting on the right of Barbara?

MEMO

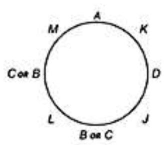
Matthew was sitting to the right of Agatha, so Kevin must have been sitting to her left side. To the left of Kevin might have been sitting either Celine or Daphne.

Lets consider both cases:

a) If it was Celine sitting to the left of Kevin, then to her left must have been John (Kevin and Matthew are already elsewhere, and Leon is Celine's husband). We know that Daphne was also sitting next to John, so she must have taken a seat to his left. The remaining two seats were occupied by Leon and Barbara (See image), which means that to the right of Barbara was Leon.



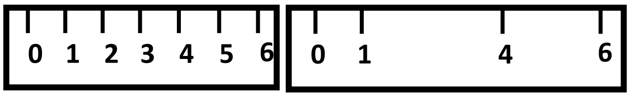
b) If it was Daphne sitting to the left of Kevin, then, per the guidelines of the problem, to her left side sat John. The next three seats were occupied by: Barbara, Celine, and Leon; Leon was sitting between Barbara and Celine (otherwise the two ladies would have been sitting next to each other) – as per picture. In that case, however, Celine would have been neighbouring on her husband, Leon. This case therefore is rejected as it violates the rules.



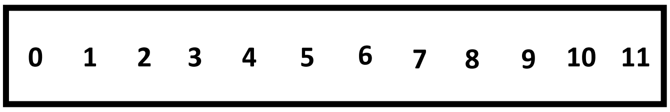
ANSWER: Leon

Question 13

If we erase 3 marks from a 6cm long ruler, and remove 3 numbers written below them (as in the picture), we will get a new ruler consisting of four marks only. Using this ruler, we will, however, still be able to measure in whole numbers each distance from 1 to 6. For example, we can measure 2cm because this distance lies between the remaining marks 4 and 6.



What maximum number of marks and numbers can we remove from an 11cm ruler, and yet be able to measure each distance from 1cm up to 11cm?



MEMO

We must leave the marks 0 and 11 in, for otherwise it would not be possible to measure out 11. We need to leave in at least another 4 marks to be able to satisfy the conditions. An Example of the ruler below.

ANSWER: 6 at most

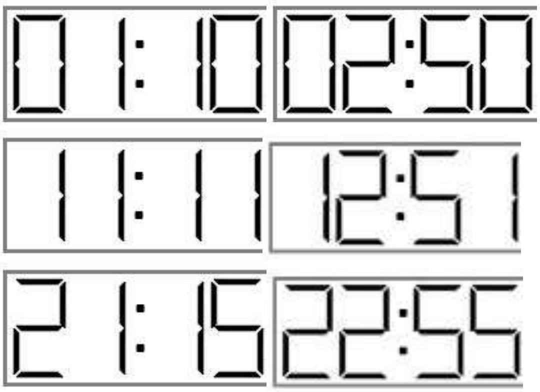
Question 14

Consider a digital clock face showing hours and minutes as below:

You place a mirror in front of the clock so that the clock face is reflected in the mirror.

How often in a 24-hour period will the reflection of the time in the mirror be the same as the actual time on the clock?
Hint: one such occasions would be at 11:11 AM.

MEMO



ANSWER: 6

Question 15	**MEMO**
Given: a three digit number abc. None of the digits is equal to 0. If $abc = a! + b! + c!$ determine the number abc	Since $7! = 5040$ is a four digit number abc cannot contain a 7. Similarly it cannot contain an 8 or 9. $6! = 720$ is a three digit number however it contains a 7, which we have summised it not allowed. Therefore it cannot contain a 6. Notice that $4! + 4! + 4! = 72$ but this is less than three digits therefore abc must contain a 5. $5! + 5! + 5! \neq 555$ therefore we can have at most 2 digits equal to 5. $5! + 5! + 1! = 241$ $5! + 5! + 2! = 242$ $5! + 5! + 3! = 246$ $5! + 5! + 4! = 264$ so we can conclude on 1 digit of 5 Now the largest sum we can make is $5! + 4! + 4! = 168$ thus we must have $a = 1$ therefore we know we have a 1 and a 5. $1! + 4! + 5! = 145$ **ANSWER: 145**

20 Mark Questions

Question 16

When Augustus de Morgan (a mathematician who was born and died in the 19th century) was asked about his age, he replied: I was x years old in the year x^2.
What year was he born in?

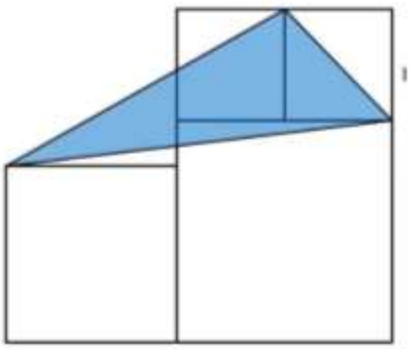
MEMO

The year x^2 must have been in the 19th century.
We can check and see that the oly square in the range of 1801 to 1900 is the number $43^2 = 1849$.
De Morgan was thus 43 years old in the year 1849, hence he was born in $1849 - 43 = 1806$.

ANSWER: 1806

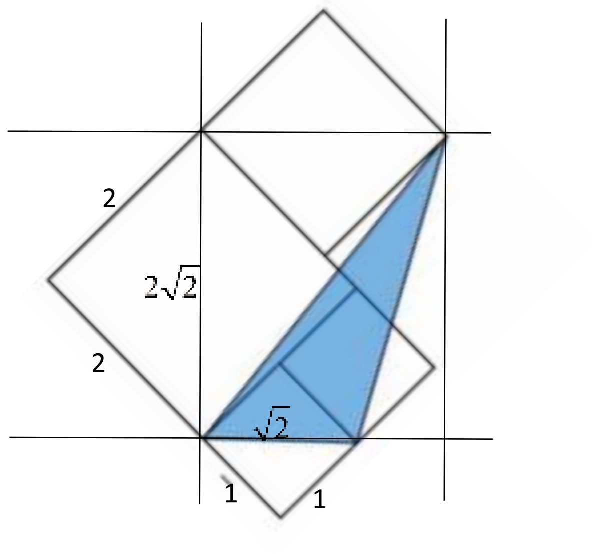
Question 17

Consider the diagram which is made up of 4 squares, the smallest of which has a side length of 1 unit.
What is the area of the triangle?



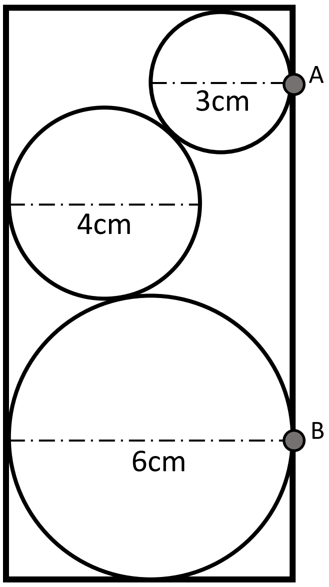
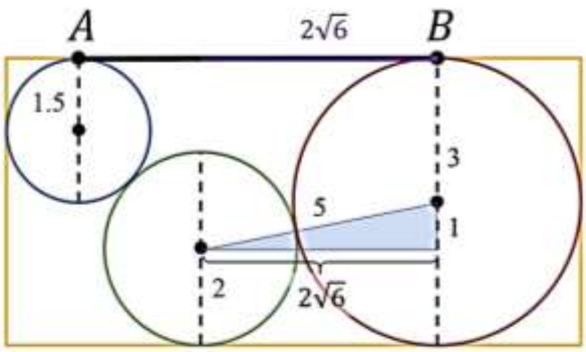
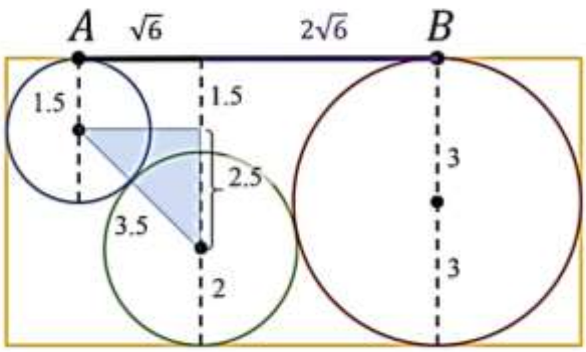
MEMO

First rotate the image for a better perspective



$$\begin{aligned}\text{Area} &= \frac{1}{2} \times \text{base} \times \textit{height}\\ &= \frac{1}{2} \times \sqrt{2} \times 2\sqrt{2}\\ &= 2\end{aligned}$$

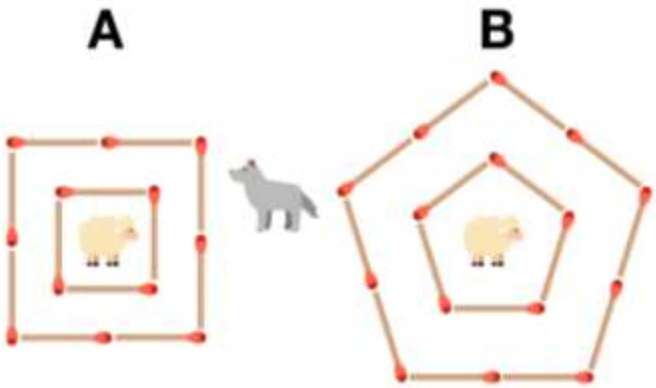
ANSWER: 2

Question 18	**MEMO**
In the diagram, thee circles of diameter 3cm, 4cm and 6cm have been placed in a rectangle as shown. What is the length of AB?  	  $AB = 2\sqrt{6} + \sqrt{6}$ $= 3\sqrt{6}$ $\approx 7{,}35cm$ **ANSWER:** $3\sqrt{6}$ **or 7,35**

Question 19

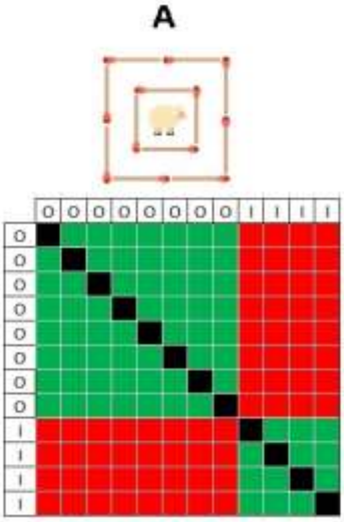
Each sheep has two fences (represented by matchsticks) protecting it from the wolf.

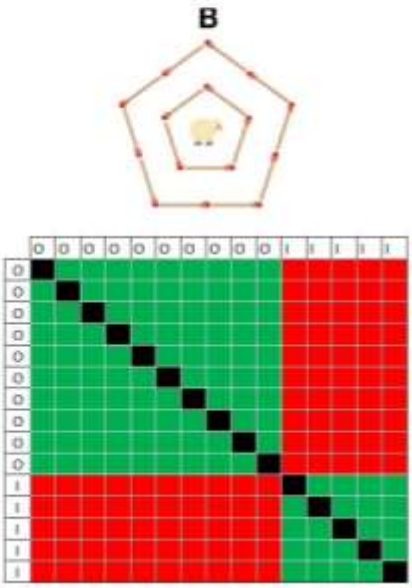
If any two matchsticks are removed at random, what is the probability of the wolf reaching the sheep?



MEMO

Creating a decision table for each:

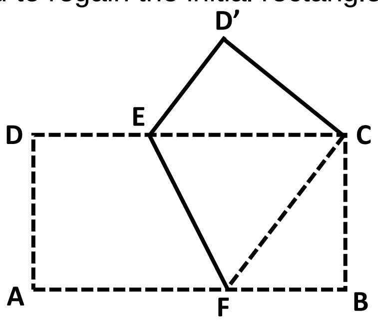




Pen	A	B	Sum
Total Outcomes	132	210	342
Safe Outcomes	68	110	178
Unsafe Outcomes	64	100	164

$$
\begin{aligned}
P(\mathrm{A} \cup \mathrm{B}) &= \mathrm{P(A)} + \mathrm{P(B)} \\
&= \frac{164}{342} \\
&= 0,4795 \\
&\approx 48\%
\end{aligned}
$$

ANSWER: 47,95%

Question 20	MEMO
A rectangular, white sheet of paper with verticies ABCD and an area of 20cm^2 was folded and pressed in such a way that its opposite vertices A and C touched each other. In this way, pentagon BCD'EF was created with an area of 12cm^2; both side were painted green, and then the paper was unfolded to regain the initial rectangle.  One side of the rectangle is now completely green and the other is two-colour. What is the area of the remaining white section?	Pentagon BCD'EF formed of a folded sheet of paper was painted bilaterally. I.e. $24cm^2$ of paper. After unfolding we have $40cm^2$ of paper (two sides measuring $20cm^2$ each). i.e., the white part takes up $40-24=16cm^2$. **ANSWER: 16**

Total: 200

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